

Name: _____ Student number: _____

IT Essentials v7.0: Introduction to Advance Computer Hardware

Chapter 3: Online Study Questions



Directions: Use the online curriculum in chapter one to answer the following questions.

Curr	Question	Answer										
3.1	Boot the Computer											
3.1.1	POST, BIOS, CMOS and UEFI											
3.1.1.1	Watch the Video	“Video Demonstration: BIOS – UEFI Menu”										
	1. True or False all Computers use the same keystroke to enter the UEFI BIOS											
	2. Complete the table of UEFI setting menu											
	<table><tr><th>Menu Item</th><th>Brief Description</th></tr><tr><td>3. This section of the UEFI to flash the BIOS</td><td></td></tr><tr><td>4. This is typically used to flash your UEFI to the latest firmware version</td><td></td></tr><tr><td>5. Data stored on CMOS (Complementary Metal Oxide Semiconductor) memory chip</td><td></td></tr><tr><td>6. If the computer’s setting including the date and time are not retained after the computer is restarted what may need to be replaced?</td><td></td></tr></table>	Menu Item	Brief Description	3. This section of the UEFI to flash the BIOS		4. This is typically used to flash your UEFI to the latest firmware version		5. Data stored on CMOS (Complementary Metal Oxide Semiconductor) memory chip		6. If the computer’s setting including the date and time are not retained after the computer is restarted what may need to be replaced?		
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3.1.1.4	7. What does UEFI and acronym for?											
	8. True or False UEFI supports the use of keyboard and mouse											
3.1.2	BIOS/UEFI Configuration											
3.1.2.1	Watch the Video	“Video Demonstration – Configure BIOS – UEFI Settings”										
3.1.2.3	9. When should you update your BIOS?											
	10. What type of memory module allows you to flash the BIOS											
	11. What are some of the improvements for updating the BIOS motherboard manufacturers are providing?											
3.2	Electrical Power											
3.2.1	Wattage and Voltage											
	12. A power supply produces several different _____											
	13. How are power supplies rated?											
	14. What law does V=IR represent?											

3.2.1.2	15. Describe the location and purpose of the voltage selector	
	16. What determines which setting to use on the voltage selector?	
3.2.	Power Fluctuation and Protection	
3.2.2.2	17. What device protects against damage from surges and spikes?	
	18. What device Supplies a consistent level of electrical power to a computer or other device and the battery is constantly recharging?	
	19. What device providing a backup battery to supply power when the incoming voltage drops below the normal level but, the battery is on standby during normal operation?	
	20. What should not be plugged into a UPS?	
3.3	Advanced Computer Functionality	
3.3.1	CPU Architectures and Operation	
3.3.1.1	21. A CPU executes instructions by following which two instruction sets?	
	22. Define Cache	
3.3.1.2	23. Define Hyperthreading	
	24. Define HyperTransport	
	25. How is the speed of a CPU measured?	
	26. The amount of data a CPU can process at one time is dependent on what?	
	27. Define Overclocking	
	28. What is Throttling and why is it used in laptops and mobile devices?	
	29. Define CPU Virtualization	
	30. If you have trouble installing a virtual machine what should you check?	

	31. Describe the NX bit	
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